

Assignment 4: CRR and BS Models, Option Pricing and Implied Volatility

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Group 42

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Question 1 – CRR European call prices

We implement the standard Cox–Ross–Rubinstein binomial tree with N time steps of length $\Delta t = T/N$, up- and down-factors

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}},$$

and risk-neutral probability accounting for the continuous dividend yield δ ,

$$q = \frac{e^{(r-\delta)\Delta t} - d}{u - d}.$$

The price at $t = 0$ is obtained by backward induction with discount factor $e^{-r\Delta t}$ per step on the payoff $V_T = (S_T - K)^+$. The implementation is in the function `EuroCRR(N)`; using the baseline parameters $S_0 = K = 100$, $r = 0.02$, $\sigma = 0.30$, $\delta = 0.01$, $T = 1$ we obtain the values reported below.

N	CRR call price
5	12.8333
10	11.9555
20	12.0993
50	12.1866
100	12.2159
250	12.2335
500	12.2393
1000	12.2423

Table 1: CRR European call prices for the baseline parameters as a function of N .

Question 2 – Black–Scholes price

For a dividend-paying stock the Black–Scholes formula for a European call is

$$C^{BS} = S_0 e^{-\delta T} \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad d_{1,2} = \frac{\ln(S_0/K) + (r - \delta \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

The analogous put formula is implemented in the function `BSPrice`. With the baseline parameters, the Black–Scholes price of the European call is $C^{BS} \approx 12.2452$.

Question 3 – Convergence plot

The figure below plots the CRR prices from Question 1 against N on a logarithmic x -axis, with the BS price overlaid as a horizontal reference line.

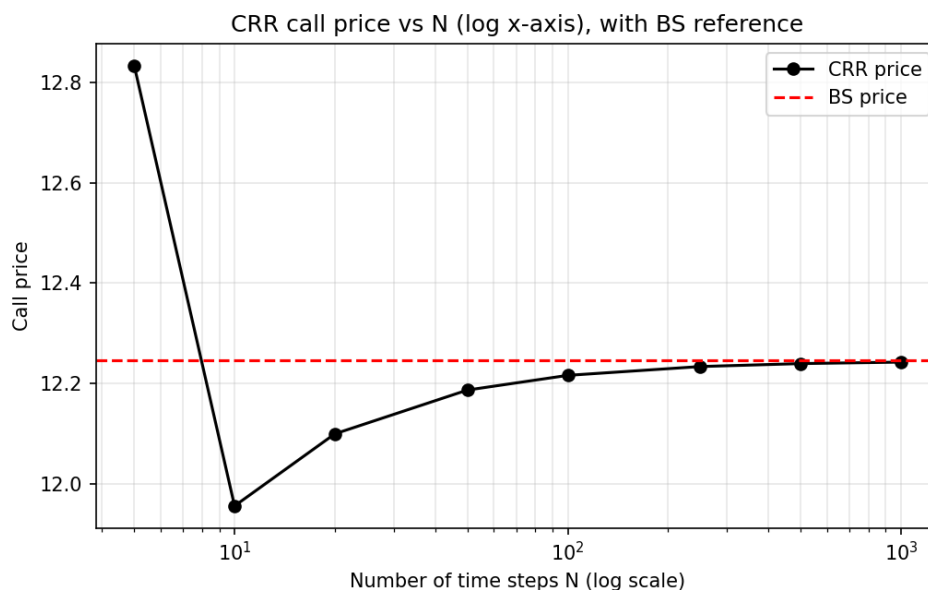


Figure 1: CRR European call price as a function of N (log scale), with the BS price as a dashed reference.

Behaviour as $N \rightarrow \infty$. The CRR prices oscillate around the BS value and converge to it as N grows: the binomial tree under the CRR parametrisation matches the first two moments of the log-return per step with the lognormal distribution assumed by Black–Scholes, so by the central limit theorem the terminal distribution under $\mathbb{Q}$ converges to a lognormal, and discounted risk-neutral expected payoffs therefore converge to the BS price. The convergence is non-monotonic (sawtooth pattern) due to the fact that whether K is hit exactly by a tree node depends on the parity of N , but the envelope shrinks at rate $O(1/N)$.

Question 4 – Digital call

The digital call pays $\mathbf{1}_{\{S_T > K\}}$ at maturity. Using the same CRR tree with $N = 100$ steps and identical u , d , q as in Question 1, we set the terminal payoff vector to 1 above strike and 0 otherwise and backward-induct using the risk-neutral probabilities and the per-step discount factor $e^{-r\Delta t}$. With the baseline parameters, the CRR price of the digital call at $N = 100$ is $C_{N=100}^{\text{digital, CRR}} \approx 0.4834$. For comparison, the closed-form Black–Scholes digital call is $e^{-rT}\Phi(d_2)$, which equals 0.4827, in good agreement with the binomial value.

Question 5 – Asian call

Comparison with European. The Asian call pays $(S^* - K)^+$ where $S^* = \frac{1}{N} \sum_{n=1}^N S_{n\Delta t}$. Averaging the underlying over N dates reduces the variance of the price input to the payoff relative to the single terminal value S_T used by the European call. Since both options are convex (call) in their respective price inputs, a less volatile input produces a smaller risk-neutral expected payoff. We therefore expect the Asian call to be *cheaper* than the otherwise-identical European call.

Path-dependence. The Asian payoff depends on the whole path of $(S_{n\Delta t})_{n=1,\dots,N}$, not only on the terminal node, so standard backward induction on the recombining tree is not valid (different paths reaching the same terminal node carry different averages). With $N = 12$ monthly steps the total number of paths is $2^{12} = 4096$, which is small enough to enumerate explicitly. For each path we compute the price path starting from S_0 (excluded from the average per the problem statement), the discrete average S^* , the path payoff, and the path probability $q^{n_u}(1-q)^{N-n_u}$ where n_u is the number of up-moves. The price is the discounted expectation

$$C^{\text{Asian}} = e^{-rT} \sum_{\text{paths } i} \text{Pr}(i) (S_i^* - K)^+.$$

With the baseline parameters and $N = 12$ we obtain $C^{\text{Asian}} \approx 7.5622$, whereas the European call with the same $N = 12$ steps prices at $C_{N=12}^{\text{Euro,CRR}} \approx 12.0032$. The Asian premium is roughly 63% of the European premium, confirming the qualitative argument above.

Question 6 – Implied volatility curve for TSLA

For each of the 75 traded TSLA options observed on 8 May 2026 (expiry 5 June 2026), we invert the Black–Scholes pricing formula numerically against the `MidPrice` to obtain the implied volatility (IV). The inversion is carried out per row via `uniroot` on $\sigma \in [10^{-6}, 5]$, using each row’s own T , S_0 , K , r , δ , and option type. The table below reports the first six rows in the order of the dataset.

Strike	Type	MidPrice	IV (%)
250	Put	0.220	82.03
265	Put	0.275	76.07
630	Call	0.450	63.49
660	Call	0.315	66.66
670	Call	0.285	67.77
690	Call	0.225	69.57

Table 2: First six options of the dataset with their recovered implied volatilities.

The figure below plots the recovered IVs (in percent) against the strike, sorted in increasing order of K .

Shape of the curve. The implied-volatility curve is U-shaped, i.e. a *volatility smile*, with a clear asymmetric tilt: the minimum lies near the at-the-money region (strike around 420–430, where IV bottoms out near 43%), while IVs rise sharply on the downside towards roughly 95% at the lowest strike $K = 220$ and more gradually on the upside towards about 70% at the highest strike $K = 690$. The left wing is steeper than the right wing, so the curve is more precisely a *volatility skew/smirk*, which is the standard pattern seen in equity index and single-name option markets.

Empirical validity of Black–Scholes. The Black–Scholes model assumes that the underlying log-returns are normally distributed with a single, constant volatility σ . If that assumption held in the data, then inverting BS at each strike would recover the same number, and the IV curve would be flat. The observed smile/skew is a direct empirical rejection of this assumption: option prices are consistent with a single constant σ only at one strike at a time, and any single σ chosen across strikes would misprice the rest of the smile. In practice the smile reflects fatter-than-lognormal tails (more probability mass on large moves, particularly on the downside for

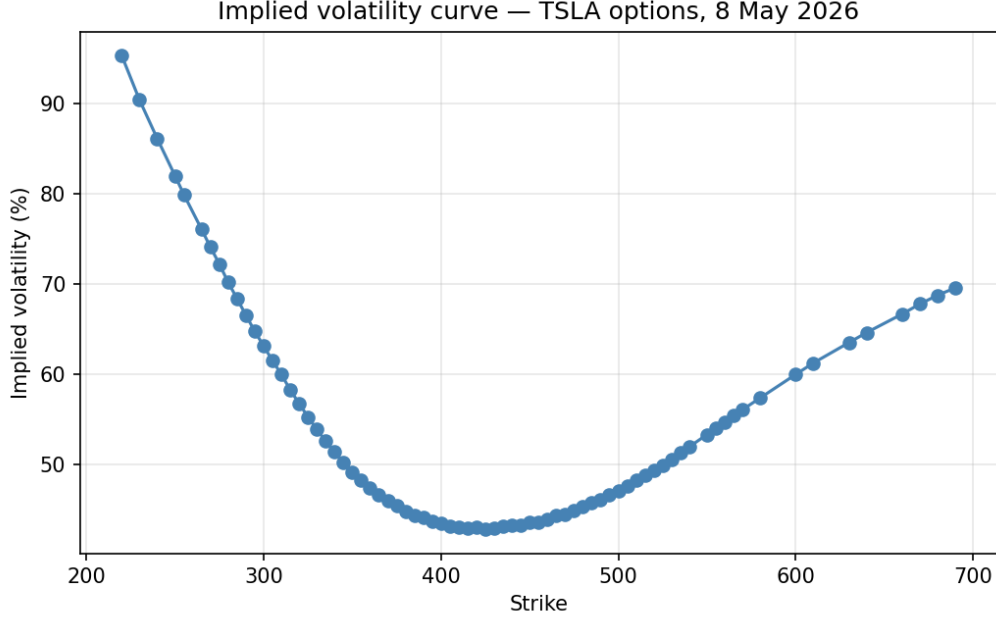


Figure 2: Implied volatility (in %) of TSLA options as a function of the strike, sorted in increasing order of strike. The spot price on the observation date is $S_0 = 427.08$.

equities) and stochastic/jump features that BS ignores. So while BS remains useful as a quoting convention and benchmark, the smile shows it is not an empirically valid *pricing* model across the full strike range.

Over-/under-pricing at the wings. Suppose we pick a single σ (e.g. at-the-money IV, $\approx 43\%$) and use BS with that σ to price all options. Because Vega is strictly positive, the BS price is monotonically increasing in σ . The market-implied volatility at the highest and lowest strikes (around 70% and 95%, respectively) is strictly larger than this single ATM σ , so the BS price computed with the ATM σ is strictly below the market price at both wings. Equivalently, BS with a constant ATM volatility *underestimates* the price at both the highest and the lowest strikes.